

# ANUP PARAJULI

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## EDUCATION

**The Ohio State University** ◇ Columbus, OH, USA Expected Graduation: 2032  
Ph.D. in Molecular Genetics | College of Arts and Sciences

**Soka University of America** ◇ Aliso Viejo, CA, USA 08/2022 – 05/2026  
Bachelor of Arts in Liberal Arts | Life Sciences Concentration GPA: 3.87/4.00

**University of Barcelona** ◇ Barcelona, Spain 01/2025 – 05/2025  
Study Abroad Program | Spanish Language Courses

## RESEARCH EXPERIENCE

***In Silico* Structural Analysis of Cancer-Associated Nit1 Mutations** 09/2025 – 05/2026

Mentor: Dr. Susan Walsh, Dr. Hung Phan ◇ Undergraduate Capstone Project ◇ Aliso Viejo, CA

- Assessment of *NIT1* mutations using in silico functional impact scoring tools (CHASMplus, Mutation Assessor, PhastCons, PROVEAN) to predict pathogenicity
- Development of a computational workflow to predict structural effects of Nit1 mutations linked to tumor suppression, integrating COSMIC mutation data with ChimeraX-based structural modeling

**Tandem Repeat Reflectin Protein Synthesis & Characterization** 06/2024 – 05/2026

Supervisor: Dr. Robert Levenson ◇ Aliso Viejo, CA

- Synthesized tandem repeat reflectin constructs using a novel optimized method
- Purified reflectin proteins and conducted phase diagram analysis to map the conditions that determine the driving forces for tunable reflectin assembly

## TEACHING & LABORATORY EMPLOYMENT

**Chemistry Laboratory Teaching Assistant** 09/2023 – 05/2024

Supervisor: Dr. Robert Levenson ◇ Aliso Viejo, CA

- Helped students operate analytical instruments including IR spectroscopy, UV/Vis spectroscopy, Microplate Reader, and Melting Point Analyzer
- Guided students through laboratory protocols, experimental design, chemical analysis, and safety procedures

## PRESENTATIONS & CONFERENCES

**American Society for Cell Biology (ASCB) National Conference** ◇ Pennsylvania, PA 2025

Poster: Nucleocytoplasmic Shuttling of the Zebrafish Sox21 Orthologs (Co-author, non-presenting)

**Soka Summer Student Research Poster Session** ◇ Aliso Viejo, CA 2025

Poster: Understanding the molecular basis of tunable reflectin assembly (Group)

**American Society for Cell Biology (ASCB) National Conference** ◇ San Diego, CA 2024

Poster: Production of Total Amino Acid Replacement Reflectin Mutants

**Soka Learning Cluster Fair 2024** ◇ Aliso Viejo, CA 2024

Poster: Antimicrobial activity of *Bitis nasicornis* and *Pseudechis colletti* (Group)

## MANUSCRIPTS

**Parajuli A**, Phan H, Walsh S. 2026. In silico protein structure analysis of nine deleterious NIT1 missense mutations identified in human cancers. *microPublication Biology*. **Submitted**

Aslam, M. H., **Parajuli, A.**, Chiriateg, I., and Levenson, R. H., Sequence Determinants of Dynamic and Reversible Reflectin Assembly. ***In preparation.***

## SKILLS

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**Molecular Genetics:** Western Blot, PCR, Cloning, Bacterial Culture, Tissue Culture, Gel Electrophoresis, Mini-prep, Nuclear Fractionation, Emulsiflex, Lyophilizer, CRISPR

**Protein Purification and Characterization:** FPLC, HPLC, DLS

**Microscopy:** Confocal Microscopy, Immunofluorescence Microscopy

**Statistical:** Regression Analysis, Hypothesis Testing, Data Cleaning, Normality Assessment, Data Transformation, and Visualization

**Chemistry Laboratory:** NMR, IR spectroscopy, UV/Vis spectroscopy, Thin Layer Chromatography

**Computational:** ChimeraX, Python (Pandas, Matplotlib, NumPy, Tkinter, NetworkX), R (dplyr, ggplot2), Jupyter Notebook, Linux Command Shell, Image Lab, MEGA 11, Benchling, Galaxy, ImageJ, ChemDraw, Microsoft Excel, L<sup>A</sup>T<sub>E</sub>X, Photoshop, Illustrator

**Additional Skills:** Creative Writing, Front-end Web Development, Video Editing, Guitar, Cooking

**Languages:** Nepali (Native), English (Fluent), Spanish (Intermediate), Hindi (Fluent)

## HONORS AND AWARDS

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**University Fellowship** | The Ohio State University 2026

- Awarded to academically outstanding students with excellent credentials

**John Stauffer Charitable Trust Summer Research Fellowship** | Soka University of America 2024

- Competitively awarded a summer research fellowship including a stipend and travel support to a national conference

**Dean's List** | Soka University of America Fall 2023 – Spring 2026

- Awarded to academically outstanding students who achieve a GPA greater than 3.75/4.00 each semester

**Soka Opportunity Grant** | Soka University of America 2022 – 2026

- Awarded a full tuition undergraduate scholarship

**Soka Merit Award** | Soka University of America 2022 – 2026

- Awarded in addition to full tuition scholarship for other academic needs

## EXTRACURRICULAR ENGAGEMENT

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**Co-President, First Gen** ◇ Aliso Viejo, CA 09/2024 – 05/2026

- Revitalized the First Gen Affinity Group and advocated for the academic and social support needs of first-generation college students
- Partnered with the Office of Financial Aid to guide peers through FAFSA and financial aid applications
- Created a responsive website using HTML, CSS, JavaScript, and CampusGroup API [<view website>](#)

**Secretary, Code Soka** ◇ Aliso Viejo, CA 09/2022 – 05/2026

- Organized weekly workshops on algorithms and coding in Python for game development and website building
- Facilitated coding projects and peer learning by debugging code, troubleshooting IT issues, and guiding software environment setup
- Created a game in Python (graphics.py) and website to reinforce biology vocabulary [<play game>](#)

**Organizer, Google Developer Groups - Soka University** ◇ Aliso Viejo, CA 09/2024 – 12/2024

- Co-organized the DevFest 2024 - Greater LA Area event (more than 350 participants)